

Mastering Tinkering assignment 5.4

Group Project - evaluation

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Design description

Prototype

The design of the wind tunnel uses a funnel shape as can be seen in figure 1. By making use of the Bernoulli principle, this funnel shape will increase the speed of the air that enters the restricted area of the wind tunnel. To draw air into the wind tunnel in the first place, a propellor attached to a 350-watt variable speed motor is used as can be seen in figure 2. This propellor in and of itself causes airflow that is turbulent. To be able to properly read the stability of the planes that are put into the wind tunnel, it is beneficial to use air with a laminar flow. To turn the turbulent airflow into a laminar airflow, a laminarization filter is used. This honeycomb filter can be seen in figure 3. The filter was to be 3d-printed, which, given the size restrictions of the print-bed, required the filter to be split up in multiple pieces.

To be able to hold the motor at the right spot in the wind tunnel, a motor mount was created. This motor mount is seen in figure 4.

To be able to give the person using the wind tunnel a view that allows them to evaluate the stability of the airplane that will be placed in the tunnel, a plexiglass panel is used as a door as can be seen in figure 5.

Finally, to enable a tinkerer to truly test their paper airplane design in the wind tunnel, a clip on a string is attached to the laminarization filter such that the plane is kept in the tunnel while allowing vertical movement. The string is kept short enough such that the airplane is kept at a slight angle to assist the airplane when it first starts to glide through the air. Simultaneously, the string should be long enough such that the airplane has the freedom of movement to show instability. This attachment mechanism can be seen in figure 6.

Design iteration

The first iteration of the motor mount was held together using wooden clamps at the top and slotted wooden pieces at the bottom. During one of the tests with the tunnel, the mount tipped forwards slightly, and the propellor ended up scratching the inside of the shroud of the tunnel. To avoid this in further instances, the bottom part of the motor mount was reinforced, and it was permanently affixed to the wind tunnel.

During another test of the wind tunnel, one of the wooden clamps of the motor mount came loose and hit the propellers. Given this occurrence, some additional safety precautions were taken in the shape of adding netting from a drone cage to the air inlet of the wind tunnel. This would ensure that the motor would not be accessible from the back of the wind tunnel, such that it is more difficult for an object or an individual's limbs or digits to enter the wind tunnel while the motor is running. In addition, this would also help to prevent any larger objects from inside the wind tunnel from being launched out. The motor is now encased on one side by the laminarization filter and on the other side by the safety netting. The safety netting can be seen in figure 7.

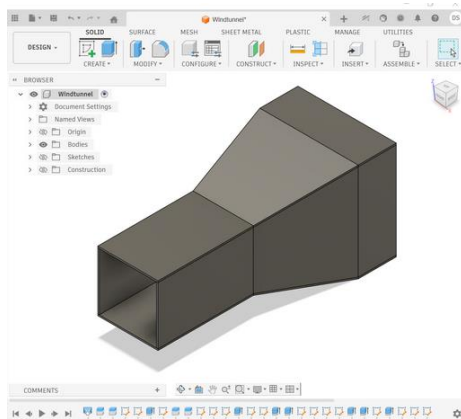


Figure 1: CAD of housing of windtunnel



Figure 2: motor and propellers

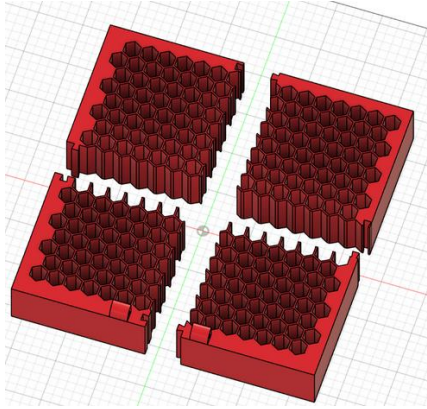


Figure 3: CAD of laminarization filter



Figure 4: motor mount



Figure 5: final prototype



Figure 6: clamp for airplane attachment



Figure 7: safety netting

Evaluation method

For the evaluation of the tinkering playground, it is important to keep in mind the goal and the process to reach that goal. To be able to evaluate these two main aspects, a choice was made to find a volunteer who could perform a tinkering session with the first iteration of the playground.

During this tinkering session, two goals were set: to make a paper airplane that was stable during flight and to learn about fundamental aerodynamics. To let the volunteer achieve this, they were supplied with materials and scaffolding while the wind tunnel was set up in the design lab.

For the material, they were supplied with multiple sheets of paper to create their own airplanes. For the scaffolding, they were supplied with a website that shows different ways to fold a paper airplane and a facilitator that would share information on how to make the paper airplane more stable.

An observer was sent along with the volunteer to be able to give an overview of the process performed by the volunteer, which could be used for the evaluation. Once the tinkering session was completed, the volunteer was taken aside for an interview related to how they experienced the session.

Only a single volunteer was chosen for the evaluation. The choice was made to be able to focus on a qualitatively high evaluation that would allow for effective iteration of the tinkering playground. Although given time constraints, no further iteration took place.

Evaluation results

Observation

The session began with a short, facilitated introduction. The facilitator walked the participant through the playground setup, explaining how the wind tunnel works and covering basic safety precautions. This onboarding kept the entry barrier low, consistent with the low floor principle, so the participant could focus on making rather than figuring out the environment. The participant then folded their first paper airplane from scratch, with the facilitator demonstrating how to secure it to the clamp inside the tunnel. Turning the motor on for the first time, the participant observed the plane under airflow and noted its behavior. From there, the participant moved into self-directed tinkering using the instruction manual as scaffolding. A second plane, built from a different template in the manual, flew noticeably lower. A third model flew high enough, but upside down entirely.

The session closed with a semi-structured interview. The facilitator asked open questions covering clarity of the instructions, whether the onboarding felt sufficient, which scaffolding materials the participant actually used and how useful they found them, what aerodynamics insights came out of the session, and whether anything was surprising.

Interview

A semi-structured interview was conducted with the participant after the playground session to evaluate mainly the performance of the playground. The main takeaway from the interview is that the playground session was engaging and accessible but limited in supporting a deeper understanding and exploration of aerodynamics.

While the participant found the instructions to start with the playground clear, there were shortcomings in the clarity of the goal and explanations about aerodynamics and how the airplane could be improved for better aerodynamics. The participant asked for clarification about the goal when asked about it in the interview; this indicates that this goal was not sufficiently communicated during the session. The participant reported a general insight during the interview, “paper airplanes are not very aerodynamic”, rather than a better understanding of aerodynamic principles.

When asked about what the participant found surprising during the session, they mentioned that the wind tunnel worked surprisingly well for the size. At the end of the interview, they also said they had a lot of fun and that it was nice to try different types of airplanes in the accessible session.

In terms of wide walls, the setup of this playground session allowed for multiple airplane designs, but exploration remained within a relatively narrow space. The participant relied primarily on existing airplane folding instructions provided as scaffolding, rather than experimenting with designs that were not provided. The narrow space for exploration is mainly due to a small range of interactions being possible with the current configuration of the wind tunnel. The opportunities for exploring a lot of different approaches or other implementations are limited.

In terms of high ceiling, the designed system had a high potential due to the complexity of aerodynamics. The participant expressed a desire for more explanation of aerodynamic principles, this is a point that can be easily improved upon. The wind tunnel provides an opportunity to let people experiment with aerodynamics in an accessible way. When proper instructions are given and more explanations about aerodynamics are provided, this playground provides a great opportunity to let people learn about aerodynamics.